

BoardNet

Board Examination Communication Network

Work Distribution Report

Group ID: 2108

This document outlines the division of work among the four group members for the BoardNet Cisco Packet Tracer project, covering topology design, addressing, routing, and all network services across the six regional examination board networks.

Required setup responsibility: G.M. Araf Mahfuz Rakib—Static and Dynamic Routing; Fatin Anjum—Dynamic Routing; Md. Safinurzzaman—Static Routing; and Md Musfiqul Islam—DHCP and Email. Therefore, every member configures at least one of the four required setups.

G.M. Araf Mahfuz Rakib		23201417
Role: Network Design, Core Routing & DNS Lead		
1	VLSM subnetting design for the base network 14.17.0.0/16 across all 6 boards and 5 inter-router links	
2	Full network topology design and interface/cable planning for all 6 routers	
3	DHK router configuration: all 5 interfaces, RIP v2 setup, static routes to CTG/BAR/KHU, static route redistribution into RIP, KHU-BAR link subnet routing fix	
4	CTG router configuration: interface addressing and default route to DHK	
5	Overall project coordination and final documentation (VLSM tree, IP addressing table, topology diagram, PDF report compilation)	
6	Central DNS configuration on CENTRAL_DNS_DHK: hostname and bare-domain A records for both web servers and all 6 mail servers, including DNS-based connectivity testing	

Fatin Anjum		22201327
Role: Dynamic Routing (RAJ & SYL)		
1	RAJ router configuration: interface addressing (LAN + central segment)	
2	SYL router configuration: interface addressing (LAN + central segment)	
3	RIP v2 configuration and verification on RAJ and SYL	
4	DHCP relay (ip helper-address) configuration on RAJ and SYL LAN interfaces	
5	Verification testing: RIP convergence, cross-board ping tests from RAJ/SYL, DHCP lease testing on RAJ and SYL LANs	

Md. Safinurzzaman		22301419
Role: Static Routing & Redundancy (KHU & BAR)		
1	KHU router configuration: interface addressing (LAN + 2 serial links)	
2	BAR router configuration: interface addressing (LAN + 2 serial links)	
3	Static routing design and configuration on KHU and BAR (primary routes via DHK)	
4	Floating static backup route configuration on KHU and BAR for KHU-BAR redundancy (administrative distance 130)	
5	Redundancy/failover testing: verified both failure directions (DHK-KHU down, DHK-BAR down) with successful automatic rerouting	

Md Musfiqul Islam		23201578
Role: Network Services (DHCP, Web, Email)		
1	DHCP pool configuration on DHK router for DHK/CTG/RAJ/SYL networks, including excluded address ranges	
2	Dedicated DHCP server configuration for KHU (DHCP_KHU) and BAR (DHCP_BAR)	
3	Web server configuration and testing: Web_DHK and Web_CTG (HTTP service and browser access verification)	
4	Email server configuration and testing: SMTP/POP3 setup and user accounts on all 6 boards, cross-board email send/receive verification	
5	Client-side network service setup and final verification: DHCP addressing checks, web access tests, email-client configuration, and service troubleshooting across all 6 boards	

All members participated jointly in Packet Tracer topology assembly, cross-board connectivity testing, and troubleshooting sessions throughout the project.